Applied Soft Computing Journal 184 (2025) 113794

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Multi-level pooling self-adaptive evolutionary multi-objective gene selection algorithm for microarray data classification

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HIGHLIGHTS

- A novel multi-level pooling self-adaptive evolutionary framework, named MOGS-MLPSAE, is proposed.
- MOGS-MLPSAE uses a Pareto-based ranking strategy to divide individuals into pools at various levels.
- MOGS-MLPSAE applies five rules per unit pool to steer individuals toward high classification accuracy.
- MOGS-MLPSAE algorithm effectively reduces the gene selection rate while achieving high classification accuracy.

ARTICLE INFO

Keywords:
Gene selection
Microarray data
Multi-level pool
Multi-objective
NSGA-II

ABSTRACT

Gene selection is a vital preprocessing technique for cancer classification using microarray data, focusing on identifying a subset of genes that achieve high classification accuracy while minimizing the gene selection rate. Despite the prevalence of multi-objective optimization algorithms in microarray data classification, balancing high classification accuracy and effective guidance in microarray classification remains challenging for most existing algorithms. This study proposes a novel multi-objective gene selection method, MOGS-MLPSAE, which utilizes multi-level pooling and self-adaptive evolutionary techniques to enhance classification accuracy. MOGS-MLPSAE employs a Pareto-based ranking pool division strategy to facilitate cross-level learning among individuals and introduces a population-biased evolutionary mechanism with five rules to drive the population toward higher classification accuracy. Compared with seven state-of-the-art multi-objective algorithms across 14 microarray datasets, MOGS-MLPSAE achieves superior performance, with classification accuracy 1.56–8.04 % higher than other algorithms and the lowest gene selection rate (average 1 %, minimum 0.01 %). This study demonstrates MOGS-MLPSAE's effectiveness in balancing classification accuracy and gene selection rate, offering a robust solution for microarray-based cancer classification.

1. Introduction

DNA microarray technology offers a powerful tool for comprehensively investigating thousands of genes simultaneously across diverse cell and tissue types, generating large datasets in fields like clinical diagnosis and drug discovery [1]. Gene expression data, which is obtained from DNA microarray experiments, plays a crucial role in cancer disease classification and detection [2,3]. However, within such datasets, a multitude of irrelevant and redundant genes (features) have been proven to be non-contributory to class prediction. These features not only increase the computational burden on learning algorithms but also undermine classification performance, further exacerbating the challenge of the 'curse of dimensionality' inherent in high-dimensional data [4]. Feature selection (FS) is a data preprocessing technique that aims to identify the most relevant features from the original feature set, thereby simplifying the classification model and effectively addressing this challenge.

Existing FS approaches generally fall into three categories based on distinct evaluation criteria: filter-based, wrapper-based, and hybrid methods [5–7]. Filter-based methods directly evaluate each feature of the original data based on their intrinsic characteristics [8,9]. Various information theory concepts, such as mutual information (MI) [10], entropy, information gain (IG) [11], and trace ratio [12], are commonly used to construct filters [13]. Filter-based methods typically require

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https://doi.org/10.1016/j.asoc.2025.113794
Received 8 July 2024; Received in revised form 15 August 2025; Accepted 25 August 2025
Available online 26 August 2025
1568-4946/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Summary of dominance-based MOOAs.

Algorithm	Strategies	Limitations	Population-Biased Evolutionary Mechanism	dynamically adjusts population size
NSGA [47]	sorts solutions into non-dominated fronts and uses crowding distance to maintain diversity in multi-objective optimization.	Treating all objectives equally	No	No
NSGA-II [48]	fast non-dominated sorting to select high-quality solutions and crowding distance to maintain diversity	Treating all objectives equally	No	No
ϵ-Dominance [49]	combines non-dominated sorting with a niching method	Parameter sensitivity prevents accuracy-focused optimization	No	No
FDGA [50]	fuzzy logic-based dominance relation to better handle the complexity and uncertainty in multi-objective decision-making	No clear accuracy priority	No	No
GrEA [51]	Partitions objective space into grids to enhance selection and diversity in many-objective optimization.	The static boundary ignores the high-accuracy solution region	No	No
ParetoCluster [52]	using Pareto dominance to optimize relevance and redundancy for each label.	Ignore the classification accuracy	No	No
PMFS [53]	Pareto-based approach to efficiently select features for multi-label classification,	bi-objective optimization ignores the importance of accuracy	No	No

fewer computational resources and exhibit greater generalizability as they operate independently of the classifier. Wrapper-based methods leverage classifier performance as the evaluation criterion for feature subsets [14–16]. The optimal feature subset in wrapper-based methods has been identified using various search techniques, including complete search [17], recursive feature elimination [18], evolutionary algorithms [19], and swarm intelligence algorithms [20]. Compared to filter-based methods, wrapper-based methods often achieve superior classification performance due to their incorporation of the classifier within the selection process. However, this incorporation may compromise generalizability and introduce substantial computational overhead, making wrapper-based methods comparatively slower. Hybrid methods [21,22] strategically integrate evaluation methodologies from both filter and wrapper techniques, seeking to simultaneously enhance classification performance and reduce computational costs. This synergistic approach has garnered significant research attention, resulting in a diverse range of hybrid FS methods.

In the nascent phases of meta-heuristic-based FS research, studies primarily focused on bolstering classification accuracy by treating FS as a single-objective optimization problem. Examples of utilized single-objective optimization algorithms included the genetic algorithm (GA) [23], differential evolution (DE) algorithm [24], whale optimization algorithm (WOA) [25], and artificial bee colony (ABC) algorithm [26]. However, FS can also be viewed as a multi-objective optimization problem (MOOP) with two crucial objectives: maximizing classification performance and minimizing the number of selected features. These objectives are often conflicting [22,27], posing a challenge for single-objective optimization algorithms, which often result in excessive feature selection or model overfitting. Multi-objective optimization algorithms (MOOAs) can circumvent these limitations by generating a comprehensive set of Pareto-optimal [28] solutions, enabling researchers to choose the most suitable feature subset from this set and enhance the model's generalization capability. Several novel multi-objective optimization algorithms have been proposed to address various engineering and real-world problems. For example, in materials science, the integration of evolutionary algorithms with neural networks has been employed to optimize the rheological behavior of non-Newtonian mixtures comprising antifreeze and nanofluids with CuO nanoparticles [29,30]. This approach has significantly contributed to predicting the critical physical properties and rheological characteristics of nanofluids [31,32]. Pandya et al. [33,34] introduced the Multi-objective Snow Ablation Optimization Algorithm (MOSAO) for solving security-constrained optimal power flow problems incorporating wind energy sources and FACTS devices. Kalita et al. [35] proposed the Multi-objective Exponential Distribution Optimizer (MOEDO), a novel math-inspired algorithm for global optimization and real-world engineering design problems. Ganesh et al. [36] developed a Decomposition-Based Multi-Objective Symbiotic Organism Search Optimization Algorithm. Khodadadi et al. [37] introduced the Multi-objective Generalized Normal Distribution Optimization (MOGNDO) algorithm for multi-objective problems. Similarly, Khodadadi et al. [38] proposed the Multi-Objective Artificial Vultures Optimization Algorithm (MOA-VOA), which has shown promising results in solving multi-objective problems. Sharma et al. [39] introduced the Non-dominated Sorting Advanced Butterfly Optimization Algorithm. Moghaddasi et al. [40] developed a multi-objective secure task offloading strategy for blockchain-enabled IoV-MEC systems using a double deep Q-network approach. Ayar et al. [41] proposed the NSICA algorithm for feature selection in arrhythmia diagnosis. Ma et al. [42] introduced the Roulette Wheel-based Level Learning Evolutionary Algorithm for feature selection of high-dimensional data. Li et al. [43] developed a recursive framework for improving the performance of multi-objective differential evolution algorithms for gene selection. These contributions highlight the continuous efforts to improve multi-objective optimization techniques across various domains.

Generally, MOOAs can be broadly categorized into three main types based on their selection mechanisms: dominance-based MOOAs, decomposition-based MOOAs, and indicator-based MOOAs [44,45]. This study specifically focuses on dominance-based MOOAs due to their unique advantages in addressing conflicting objectives in the MOOP and their superior convergence properties, particularly in high-dimensional

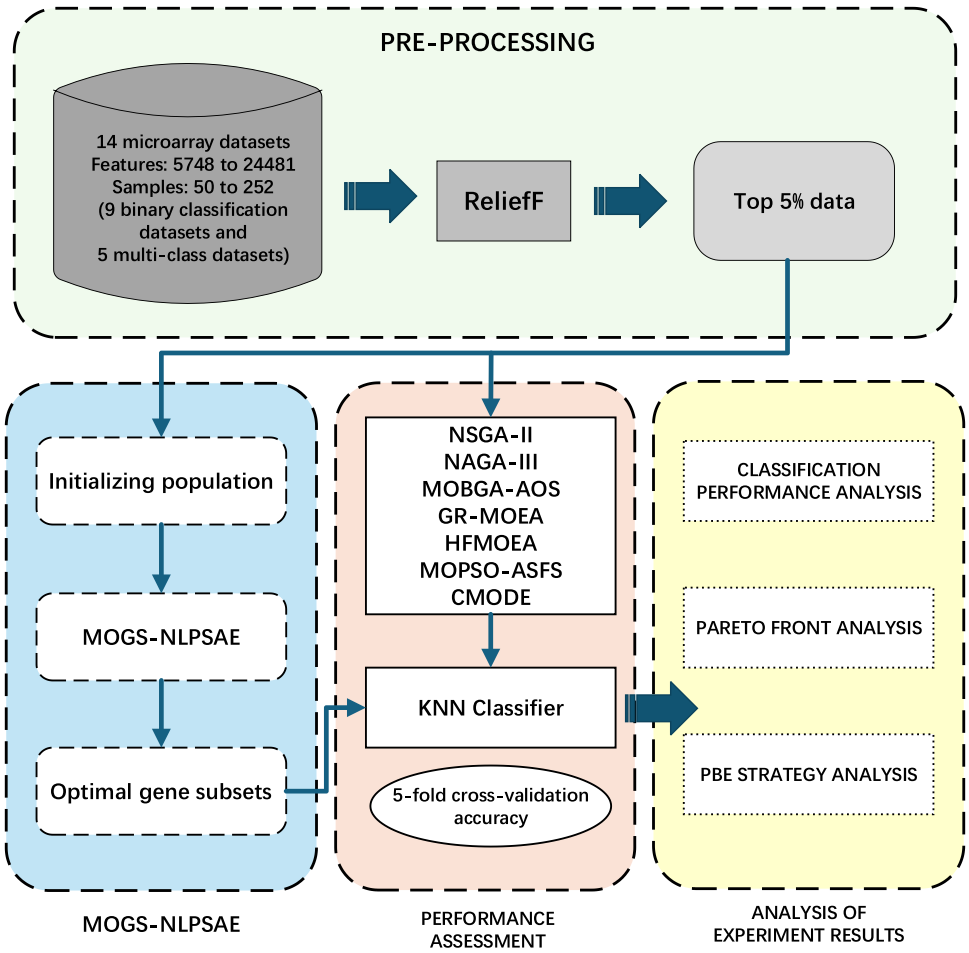


Fig. 1. Framework of the study.

FS tasks. The foundation for dominance-based MOOAs was laid by Goldberg et al. in 1989. They introduced the use of Pareto dominance relationships to identify all Pareto-optimal solutions within the population, thereby pioneering this significant advancement in multi-objective evolutionary algorithms (MOEAs) [46]. Subsequently, Srinivas et al. proposed the Non-dominated Sorting Genetic Algorithm (NSGA), which enhanced population selection through a non-dominated sorting mechanism [47]. To further address the challenges in parameter settings of NSGA, Deb et al. developed the Fast Elitist Non-dominated Sorting Genetic Algorithm (NSGA-II), which stands as a significant and influential advancement in dominance-based multi-objective optimization algorithms [48].

To further enhance the efficiency and applicability of dominance-based MOOAs, researchers have introduced various innovative dominance rules. For example, the ϵ-dominance method utilizes an ϵ parameter to regulate solution diversity and distribution [49]; however, it still faces challenges in computational complexity, parameter selection, and scalability to complex problems. The Fuzzy-Dominance method employs fuzzy logic theory to enhance the algorithm's flexibility in handling imprecise and uncertain problems [50]. The Grid-based dominance method applies grid partitioning techniques to improve performance in high-dimensional search spaces [51]. These advancements significantly strengthen the performance of dominance-based MOOAs across diverse application scenarios. In feature selection, these innovative dominance rules have been effectively applied to enhance both accuracy and efficiency. Kashef et al. developed a rapid and accurate multi-label FS method that leverages Pareto dominance to identify the most significant features for each label, achieving high precision and efficiency [52]. Furthermore, Hashemi et al. introduced a novel method that frames the multi-label FS problem as a bi-objective optimization task, evaluating feature relevance and redundancy based on Pareto dominance. This method optimizes feature selection within the bi-objective domain, significantly enhancing the effectiveness of dominance-based MOOAs in high-dimensional FS [53]. Table 1 summarizes the characteristics and limitations of the relevant dominance-based algorithms.

Although existing MOOAs excel at balancing trade-offs among multiple objectives, in FS-based classification applications, achieving high classification accuracy often takes precedence over other objectives. Traditional MOOAs struggle to effectively manage the varying importance of objectives, increasing the complexity of feature selection. To address these challenges, this study proposes a new dominance-based MOOA that focuses on enhancing classification accuracy to optimize the feature selection process and improve overall classifier performance. To achieve this goal, we introduce a novel Pareto-biased guide-based multi-objective gene selection method, named MOGS-MLPSAE. This method effectively guarantees the diversity of Pareto solutions and achieves the selection of feature subsets with high classification accuracy in high-dimensional microarray data. The process involves the following steps: First, the ReliefF algorithm is employed to effectively eliminate a substantial number of redundant and irrelevant features. Subsequently, an initial population is generated based on the reduced feature space. Then, the proposed Pareto-based ranking pool division strategy is utilized to allocate individuals from the entire population into distinct ranking pools on each layer of the Pareto front. Finally, a population evolution mechanism is employed to precisely guide the evolution process within these ranking pools, thereby directing the entire population toward achieving higher classification accuracy. Given that

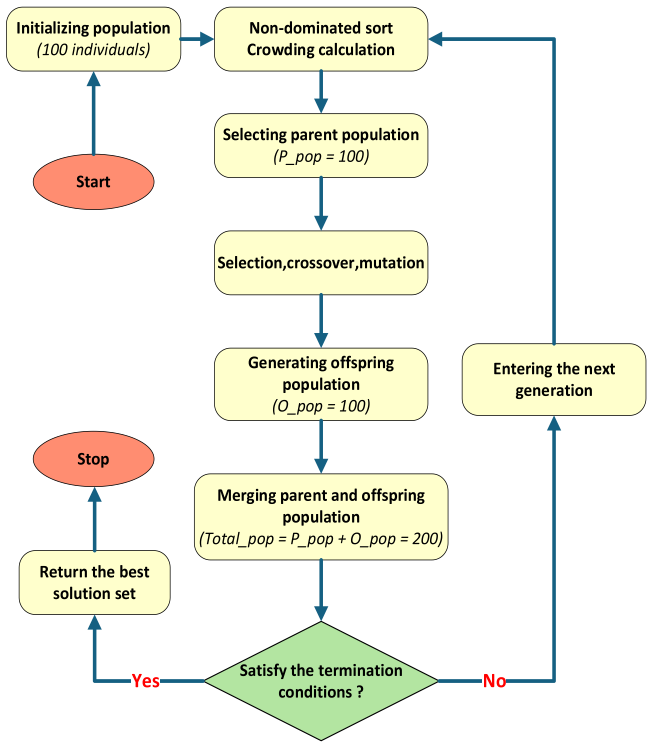


Fig. 2. Flowchart of NSGA-II (*P_pop* denotes the number of parent population individuals, *O_pop* denotes the number of offspring population individuals, *Total_pop* denotes the number of total population individuals).

each parent individual resides in a distinct unit pool environment, it follows that the number of offspring produced by each parent individual can vary. Consequently, the entire population evolves under the introduced adaptive evolution mechanism. The primary contributions of this study are summarized as follows:

1) **Novel Framework for Enhanced Classification Accuracy**: We introduce a novel multi-level pooling self-adaptive evolutionary framework, named MOGS-MLPSAE, which prioritizes individuals with high classification accuracy. This framework effectively addresses the limitations of existing multi-objective FS algorithms, which often neglect the prioritization of high accuracy while balancing two objectives: minimizing classification error and minimizing the number of selected features.
2) **Pareto-Based Ranking Pool Division Strategy**: MOGS-MLPSAE employs a Pareto-based ranking pool division strategy to allocate individuals to unit pools at different levels. This strategy facilitates cross-level learning among individuals and enhances the influence of high-accuracy individuals on the overall population.
3) **Population-Biased Evolutionary Mechanism:** Building on the ranking pools, we propose a novel population-biased evolutionary mechanism that incorporates five rules to regulate the evolutionary behavior of individuals within each unit pool. This mechanism effectively guides the entire population toward achieving higher classification accuracy.
4) **Self-Adaptive Evolution Mechanism**: To maintain a balanced population, MOGS-MLPSAE incorporates a self-adaptive evolution mechanism that dynamically adjusts the number of parent and offspring individuals. This ensures that the population remains diverse and robust throughout the evolutionary process.
5) **Superior Performance in Gene Selection and Classification**: MOGS-MLPSAE achieves a significant reduction in the gene selection rate while maintaining high classification accuracy. Extensive experiments on 14 microarray expression datasets demonstrate its superiority over seven state-of-the-art multi-objective algorithms in terms of both classification accuracy and gene set size. These results highlight its effectiveness in cancer classification.

Fig. 1 illustrates the framework of this study, which comprises the following steps: (1) Redundant and irrelevant features are eliminated using the ReliefF algorithm; (2) An initial population is generated based on the reduced feature space, followed by the generation of an optimal subset via MOGS-MLPSAE; (3) Seven competitive multi-objective feature selection (FS) algorithms are selected to benchmark the performance of the proposed MOGS-MLPSAE; (4) The effectiveness of MOGS-MLPSAE in cancer classification is demonstrated through classification performance analysis, Pareto front analysis, and PBE strategy analysis.

The remainder of the article is organized as follows: Section 2 describes the related work in this study. Section 3 presents our proposed algorithm in detail. The experimental setup, results, and analysis are presented in Section 4. Finally, the conclusion is drawn in Section 5.

2. Related works

2.1. Multi-objective optimization

Multi-objective optimization pertains to the simultaneous optimization of multiple objectives, a common problem encountered in various real-world domains including engineering, economics, and environmental management. Without loss of generality, the mathematical formulation of a MOOP is given as follows [54]:

$$\min \quad F(X) = (f_1(X), f_2(X), \ldots, f_m(X)), \quad X \in \Omega \subseteq R^n \tag{1}$$

where $X = (x_1, x_2, \ldots, x_n)$ is an n-dimensional variable vector in the search space Ω, m is the number of objectives, and $f_i(X)$ for $i = 1, 2, \ldots, m$ represents the objective function for the ith objective. The goal of a MOOP is to identify an optimal solution vector X that optimizes all the objective functions, subject to the constraints. However, due to inherent conflicts among these objective functions, it becomes unattainable to find a solution that simultaneously optimizes all of them. Instead, the objective of MOOP is typically to find a set of Pareto-optimal solutions. For two individuals X_1 and X_2, if X_1 is not inferior to X_2 in all objective functions and performs better in at least one objective function, then X_1 is said to dominate X_2 ($X_1 \prec X_2$), which is represented by Eq. 2 [55].

$$X_1 \prec X_2 \Longleftrightarrow \forall i \in \{1, \ldots, m\}, f_i(X_1) \leq f_i(X_2) \textbf{\textit{and}} \exists j \in \{1, \ldots, m\}, f_j(X_1) < f_j(X_2) \tag{2}$$

To summarize, X^* is a Pareto optimal solution if there are no solutions dominate X^*, and all Pareto optimal solutions that are mutually non-dominated constitute the Pareto front.

Let $f(X)$ represent two objectives commonly employed in other multi-objective FS [56]: (1) the number of selected features; and (2) the classification error rate associated with the feature subset. Concretely, the number of selected features $f_1(X)$ is calculated according to Eq.3. The classification error rate $f_2(X)$ is calculated in this study using the K-nearest neighbor (KNN) classifier with five-fold cross-validation and is given by Eq.4.

$$f_1(X_i) = \sum_{j=1}^{n} f_{ij}^{'} \& \tag{3}$$

$$f_2(X_i) = \frac{1}{n} \sum_{j=1}^{n} \frac{N_{err}}{N} \& \tag{4}$$

Where X_i represents the ith individual in the population, while $f_{ij}^{'}$ takes values form $\{0, 1\}$, where 1 indicates that the jth feature is selected and 0 implies that the jth feature is not selected; N_{err} represents the number of samples misclassified by the classifier, and N represents the total number of samples.

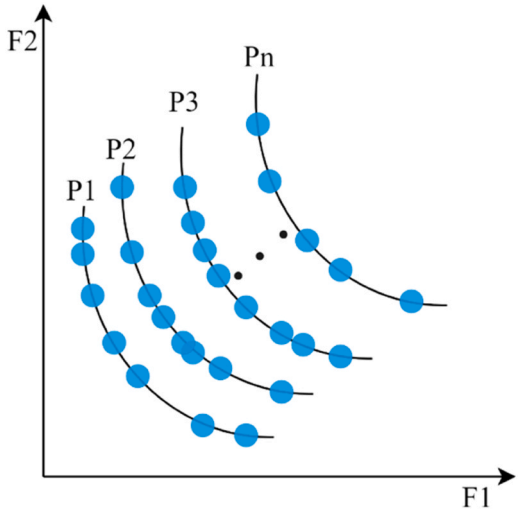


Fig. 3. Population distribution on the Pareto front. (F1 represents the gene selection rate and F2 corresponds to classification error rate).

2.2. NSGA-II

The NSGA-II algorithm, proposed by Deb et al. in 2002 [48], is a well-established technique for multi-objective optimization. It employs non-dominated sorting and crowding measures to maintain the diversity of the population and preserve the quality of the Pareto front. The overall evolution process consists of six steps: 1) population initialization; 2) calculation of the non-dominance hierarchy for each individual in the population; 3) selection of parents based on the crowding distance calculation; 4) generation of offspring using selection, crossover, and mutation operators; 5) merging parent and offspring populations into the next generation; and 6) repetition of Steps 2–5 until the termination condition is met. The flowchart of the NSGA-II algorithm, depicted in Fig. 2, showcases its implementation on a population comprising 100 individuals.

Specifically, the NSGA-II algorithm consists of five essential components, as follows:

a) Non-dominated sorting is a recursive algorithm that arranges all individuals based on the non-dominated relationship defined in Eq.2. The process initiates with the entire population and iteratively identifies individuals on the Pareto front. This selection process is repeated for the remaining individuals until all individuals have been assigned to a specific Pareto front.
b) Crowding distance is a metric that quantifies the density of individuals in the Pareto front, aiding the algorithm to select higher-quality solutions from the Pareto front.
c) The selection operator is employed to choose superior parent individuals for the purpose of generating offspring. Commonly utilized methods include roulette wheel selection [57] and binary tournament selection.
d) The crossover operator refers to the exchange of specific genes between two individuals, serving as the primary mechanism through which NSGA-II generates new individuals and enhances population diversity. Commonly employed methods include single-point crossover or multi-point crossover.
e) The mutation operator involves the stochastic modification of specific genes within an individual to generate new individuals, thereby aiding NSGA-II in circumventing local optima.

2.3. Self-adaptive mechanism for feature selection

In recent years, the self-adaptive mechanism has been proven to be an effective strategy for addressing FS problems. For example, Zhang et al. [58] proposed a self-adaptive firefly algorithm for FS that dynamically adjusts key parameters based on firefly distance, thereby achieving dynamic adaptation of the light absorption coefficient. Han et al. [59] proposed a Multi-Objective Particle Swarm Optimization FS algorithm based on adaptive strategies (MOPSO-ASFS). This method employs an adaptive penalty mechanism and an adaptive leader particle selection strategy to enhance the selection pressure of particles, thereby improving the convergence and diversity of the algorithm. Wang et al. [60] proposed a self-adaptive weighted differential evolution algorithm (SaWDE) for solving large-scale FS problems. This method designs a new adaptive mechanism that autonomously selects strategies and learns different features of the dataset based on historical information to improve the algorithm's adaptability. Xue et al. [57] introduced a self-adaptive MOEA for FS with dynamic crossover operator selection. The method integrates five crossover operators and adaptively selects the crossover operator based on the performance during the evolutionary process, improving the algorithm's performance. Wang et al. [61] proposed an FS method based on the Adaptive Balanced Grey Wolf Optimization (ABGWO) algorithm. This method introduces an adaptive coefficient to dynamically adjust the exploration and exploitation capabilities of different optimization stages, improving the algorithm's robustness and global search capability. Sun et al. [62] proposed an adaptive dual-graph multi-label feature selection algorithm (MFS-ADGO). This algorithm can learn the data geometric structure and adaptively select discriminative feature subsets, which effectively improves the performance of multi-label learning.

In conclusion, the plethora of self-adaptive mechanisms presented above underscores their effectiveness in tackling feature selection problems. By dynamically adapting parameters, operators, and strategies, these approaches overcome the limitations of static algorithms and achieve better performance, particularly for large-scale datasets. This motivated our attempt to develop a more adaptable solution for feature selection.

3. Proposed method

In this section, a multi-objective optimization algorithm named MOGS-MLPSAE is proposed based on a multi-rule ranking pool and self-adaptive evolution. MOGS-MLPSAE represents an enhanced version of the NSGA-II algorithm. Firstly, by eliminating the crowding distance calculation, the selection of parent individuals is now determined through a novel Pareto-based ranking pool division strategy (PRPD). This strategy allocates all individuals on the Pareto front into distinct levels of unit pools. Secondly, a population-biased evolutionary strategy (PBE) based on PRPD is introduced to guide the ranking pools toward learning from higher-rank pools, thereby driving the entire population toward achieving high classification accuracy. In this strategy, five new rules are designed to replace the traditional selection, crossover, and mutation operators in the NSGA-II algorithm. Finally, the proposed population self-adaptive evolution mechanism (PAE) replaces the traditional evolution mechanism of NSGA-II and effectively balances the number of parent and offspring individuals in each evolutionary process.

Next, we sequentially introduce the PRPD, PBE, and PAE mechanisms, which together form the fundamental basis for the comprehensive pseudocode of the MOGS-MLPSAE algorithm.

3.1. Pareto-based ranking pool division (PRPD)

Traditional FS Pareto-based MOOAs determine the non-dominated relationships among individuals based on their performance in two objectives: classification accuracy and the number of selected features. In the initial stages of these algorithms, a non-dominated sorting approach is employed to assign each individual to its corresponding layer on the Pareto front based on Eq.2. The sorting results are visualized through the Pareto chart shown in Fig. 3. Where, *Pn* represents the *n*th

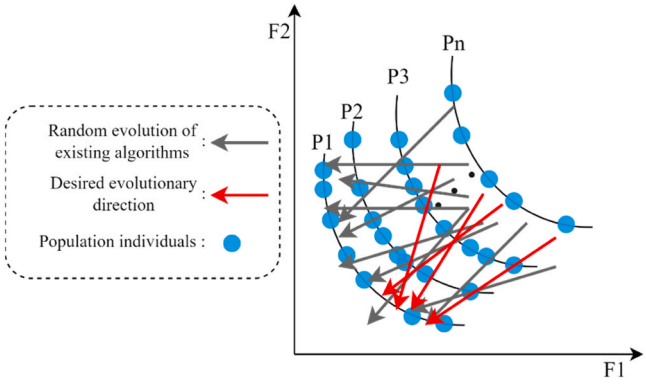


Fig. 4. Evolutionary directions on the Pareto front (F1 represents the gene selection rate and F2 corresponds to classification error rate).

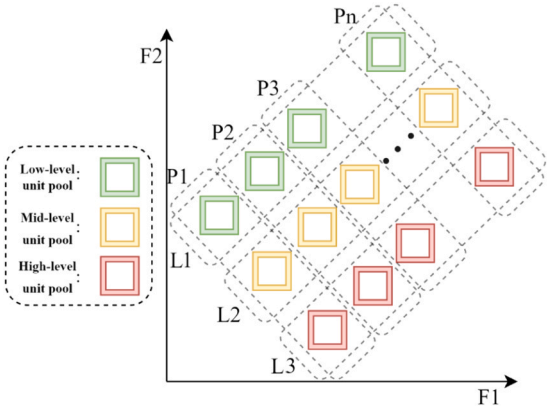


Fig. 5. Pareto-based ranking pool division (F1 represents the gene selection rate and F2 corresponds to classification error rate).

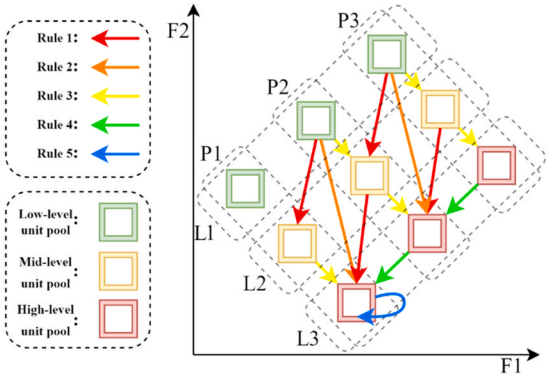


Fig. 6. The schematic diagram of PBE (F1 represents the gene selection rate and F2 corresponds to classification error rate).

layer Pareto front. Individuals on the nth Pareto front dominate all individuals on the n'th ($n' > n$) Pareto front. Individuals located on the Pareto front are regarded as dominant individuals and are selected as parent individuals for generating offspring and guiding the evolution of the population. However, in subsequent evolutionary stages, existing algorithms fail to consider an individual's Pareto rank, thereby neglecting the differences in guiding positions among individuals across different levels of Pareto fronts.

Individuals on the same Pareto front can be categorized into three groups—Class A, Class B, and Class C—based on their performance on different objectives. Class A individuals demonstrate superior classification accuracy but an inferior feature selection rate compared to other individuals. Conversely, Class B individuals exhibit lower classification accuracy but excel in feature selection rate. Lastly, Class C individuals do not exhibit any significant advantage or disadvantage in either classification accuracy or feature selection rate. Considering that the primary objective in gene selection is to maximize classification accuracy, individuals belonging to Class A hold paramount importance among the three categories, as they offer invaluable guidance for population evolution. In contrast, individuals from Class C have the potential to misdirect the evolutionary trajectory of the population, thereby impeding optimal progress. However, most Pareto-based MOOAs tend to primarily select individuals from the first few layers of the Pareto front as parent individuals to guide the population's evolution. This approach presents a challenge in mitigating the influence of individuals in Class C on improving algorithmic classification performance. The evolutionary direction of most existing Pareto-based MOOAs on the Pareto front, as depicted in Fig. 4, appears random and is represented by the gray arrows. This randomness emerges under the joint guidance of individuals belonging to Class A, Class B, and Class C. However, it is crucial for a population to strive toward evolving in the direction indicated by the red arrows, which offers greater benefits for enhancing the algorithm's classification performance.

To address the issue, a Pareto-based ranking pool division strategy (PRPD) is proposed, as depicted in Fig. 5. This strategy involves allocating all individuals of the population to different layers of the Pareto front through non-dominated sorting. Subsequently, individuals within each layer are categorized into three classes: Class A, Class B, and Class C. These classes are respectively assigned to three levels of unit pools: high-level, mid-level, and low-level unit pools. Since these three categories of individuals exhibit opposite trends in the two metrics (classification accuracy and number of selected features), we can employ one metric (classification accuracy) to classify the individuals. Specifically, we utilize the one-third and two-thirds quantiles of the classification accuracy rankings of the individuals on each layer of the Pareto front to divide them into the three levels of unit pools. This strategy allocates individuals from all layers of the Pareto front to distinct levels of unit pools, thereby facilitating more precise guidance of their evolutionary trajectory toward specific directions.

Ranking pool division algorithm is represented in Algorithm 1. Firstly, an empty ranking pool *RP* is initialized (line 1). Secondly, the classification accuracy (*accList*) of individuals on each layer of the Pareto front is calculated, and the *Value*1 (1/3 quantile) and *Value*2 (2/3 quantile) are identified (lines 3–4). Then, the individual's belongingness to a specific level of the unit pool is determined, and the individual is placed into the overall ranking pool *RP* (lines 5–13). Finally, the partitioned ranking pool *RP* is returned (line 15).

Algorithm 1. Ranking pool division

Input: PRL (Pareto ranking list with p layer Pareto front);
Output: RP (the ranking pool);
initialize $RP[rp_i] = \{\{\}, \{\}, \{\}\}$ for $i = 1$ to p;
for $i = 1$ **to** p **do**
  calculate the classification accuracy $accList$ of individuals on the i-layer Pareto front from the PRL;
  identify the one-third quantile $Vaule1$ and two-thirds quantile $Vaule2$ of $accList$;
  for each individual d_j in i-layer Pareto front **do**
    if $accList_{d_j} < Vaule1$ **then**
      put individual d_j into the low-level unit pool $RP[rp_{i,1}]$;
    else if $accList_i \geq Vaule1$ *and* $accList_i \leq Vaule2$
      put individual d_j into the mid-level unit pool $RP[rp_{i,2}]$;
    else
      put individual d_j into the high-level unit pool $RP[rp_{i,3}]$;
    end
  end
end
Return: RP;

3.2. Population-biased evolutionary (PBE)

Based on PRPD, all individuals of the population are assigned to ranking pools, where each individual is allocated its corresponding Pareto front and unit pool. To enhance the guidance of individual evolution and steer the entire population toward achieving higher classification accuracy more effectively, we propose a mechanism called population-biased evolution (PBE). The schematic diagram of this mechanism is illustrated in Fig. 6.

In Fig. 6, the five colored arrows correspond to the five newly introduced rules (Eqs. 5–9), each imposing specific constraints on the evolutionary behavior of individuals. These rules are meticulously designed to guide the population's evolution, replacing the traditional selection and crossover operators. Despite their differences, these rules share a unified objective: to steer the population toward the desired outcome —high classification accuracy— under varying conditions. Each rule is tailored to address specific individual conditions and does not hold significant standalone value. Instead, they collectively aim to achieve a synergistic effect, driving the overall evolutionary trajectory of the population toward the intended direction. In addition, each arrow originates from unit pool A, which represents the learning requirement, and terminates at unit pool B, symbolizing the learning target for unit pool A. Compared with Fig. 4, the evolution direction of the population becomes clearer, thereby increasing its potential to explore solutions with high classification accuracy. The five rules are mathematically represented by the following formulas.

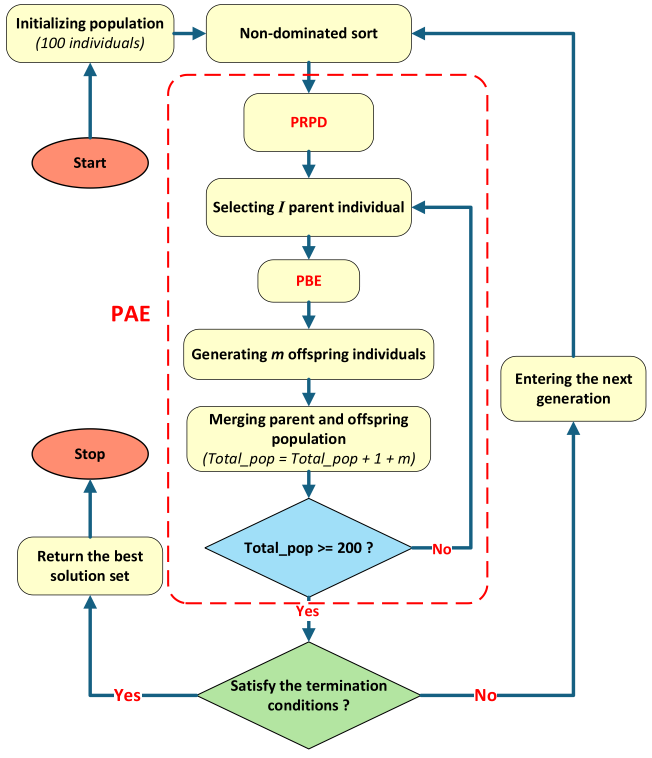


Fig. 7. Flowchart of an enhanced NSGA-II (with 100 individuals) based on PAE (*Total_pop* denotes the number of total population individuals).

$$unitB_{(p',l')} = unitA_{(p-1,\ \ l+1)} \quad (p > 1, \quad l < 3) \tag{5}$$

$$unitB_{(p',l')} = unitA_{(p-1,\ \ l+2)} \quad (p > 1, \quad l < 2) \tag{6}$$

$$unitB_{(p',l')} = unitA_{(p,\ \ l+1)} \quad (p = 1, \quad l = 2 or p > 1, l < 3) \tag{7}$$

$$unitB_{(p',l')} = unitA_{(p-1,\ \ l)} \quad (p > 1, \quad l = 3) \tag{8}$$

$$unitB_{(p',l)} = unitA_{(p,\ \ l)} \quad (p = 1, \quad l = 3) \tag{9}$$

where p represents the pth Pareto layer, l represents the lth level of unit pool, *unitA* represents the unit pool A, and *unitB* represents the unit pool B.

During the evolution of the population, each unit pool is sequentially utilized as the unit pool A, following the order of Pareto front rank first and then unit pool rank. Subsequently, unit pool A determines its learning target (unit pool B) based on the rules (Eqs.5–9) it satisfies. All individuals in unit pool A will randomly select one individual from unit pool B for learning purposes. To ensure biased evolution of the population while improving diversity, a small number of features from individuals in unit pool A are also selected by individuals in unit pool B for self-mutation during the learning process. It should be noted that learning of different-level unit pools is guided by distinct crossover rates, taking into account the varying number of valuable features in individuals from lower-level unit pools. Therefore, a high crossover rate ($CR1 = 0.9$), commonly adopted in most literature [22,57], is employed

Table 2
The basic information of datasets.

Dataset	No. of genes	No. of samples	No. of classes
TOX_171	5748	171	4
Leukemia	7129	71	2
Arcene	10000	199	2
Brain_Tumor	10367	50	4
CLLSUB	11340	110	3
11_Tumor	12533	174	11
LungCancer	12533	180	2
Prostate	12600	101	2
Ovarian	15154	252	2
SMKCAN187	19993	186	2
BreastCancer2	22283	103	2
CrohnDisease	22283	126	3
GLI	22283	84	2
BreastCancer1	24481	96	2

by individuals to acquire knowledge from the high-level unit pool. Crossover rates $CR2$ and $CR3$ are obtained with a decay factor of $\alpha = 0.15$ and are respectively utilized by individuals to learn the mid-level and low-level unit pools.

Algorithm 2. **Population-biased evolution**

Input: PI (a parent individual); RP (the ranking pool); $CR = \{0.9, 0.75, 0.6\}$ (crossover rates);
Output: OIL (Offspring individuals list);
1 find the unit pool $unitA$ that PI belongs to from RP;
2 find all unit pools $unitBList$ that can be learned by $unitA$ according to Eqs.5-9;
3 **for** $i = 1$ **to** $len(unitBList)$ **do**
4 $unitB = unitBList_i$;
5 randomly select an individual BI from the unit pool $unitB$;
6 $AI = PI$ // Duplicate PI individual;
7 identify the unit pool level $l1$ that individual BI belongs to;
8 determine the features $indexF_1$ to be exchanged based on the crossover rate CR_{l1};
9 $PI[indexF_1] = BI[indexF_1]$;
10 identify the unit pool level $l2$ that individual AI belongs to;
11 determine the features $indexF_2$ to be exchanged based on the crossover rate CR_{l2};
12 $BI[indexF_2] = AI[indexF_2]$;
13 save PI and BI individuals to OIL;
14 **end**
15 **Return:** OIL;

The detailed process of population-biased evolution is described in Algorithm 2. Firstly, the unit pool A (*unitA*) to which the parent individual (*PI*) belongs is identified (line 1). Secondly, all target unit pools (*unitBList*) for unit pool A are determined according to Eqs.5–9 (line 2). Thirdly, one individual (*BI*) is randomly selected from each target unit pool for learning purposes (line 5). Fourthly, mutual learning occurs between individuals *PI* and *BI*, with the proportion of features to be exchanged being determined by the rank of the other party's unit pool (lines 6–12). Finally, the resulting new individuals are added to the offspring individuals list (*OIL*) (line 13).

3.3. Population self-adaptive evolution (PAE)

Due to the different rules that individuals in different unit pools must meet, the number of target unit pools that can be learned is uncertain, as is the quantity of offspring individuals generated during individual evolution. To ensure an adequate number of individuals progress to the next generation in each population evolution, we propose a population self-adaptive evolution (PAE) mechanism based on the proposed PRPD and PBE. This mechanism assists the population in automatically balancing the number of parent and offspring individuals, thereby enhancing the algorithm's robustness.

Based on the PAE mechanism, the flowchart of an enhanced NSGA-II algorithm (with 100 individuals in the population) is illustrated in Fig. 7. In comparison to Fig. 2, besides incorporating the PRPD and PBE strategies, the enhanced NSGA-II algorithm adopts a single-parent selection approach and allows for non-uniform production of offspring individuals by each parent individual. It is important to note that traditional mutation strategy have been retained for its ability to increase population diversity. Additionally, the PAE mechanism assesses whether more parent individuals are necessary based on the number of generated offspring individuals and the current population size.

3.4. Pseudocode of MOGS-MLPSAE

Algorithm 4. presents the comprehensive pseudocode of the proposed MOGS-MLPSAE. The following is a detailed explanation:

1) **Feature Reduction**: The search space ($D_{s\times t'}$) is reduced by retaining the top 5 % of features using ReliefF (line 1). The ReliefF method, a widely adopted filter technique [59,63], effectively aids algorithms in eliminating numerous irrelevant features.
2) **Population Initialization**: The population *P* is initialized in a lower-dimensional (dim) feature space ($D_{s\times t'}$) according to Algorithm 3 (line 2). Since previous research [28,64] has indicated that the optimal gene subset often consists of a small number of genes, initializing individuals in a lower-dimensional space incorporates essential information related to the optimal gene subset.
3) **Ranking Pool Division**: The ranking pool *RP* is divided using non-dominated sorting and the PRPD strategy (lines 5–6).
4) **Parent Selection and Offspring Generation**: Select a parent individual (*PI*) in sequence following the order of Pareto front rank first and unit pool rank later. Employ the PBE mechanism and the traditional mutation operator to generate offspring individuals (*OIL*) (lines 10–11).

Table 3
Parameter settings in all comparison algorithms.

Algorithm	Parameter	Value
NSGA-II [48]	crossover rate	$CR = 0.9$
	mutation rate	$MR = 1/D$
NSGA-III [65]	crossover rate	$CR = 0.9$
	mutation rate	$MR = 1/D$
	reference point number	$Pnum = 4$
MOBGA-AOS [57]	crossover rate	$CR = 0.9$
	mutation rate	$MR = 1/D$
	number of repeated	$LP = 5$
GR-MOEA [66]	crossover rate	$CR = 0.9$
	mutation rate	$MR = 1/D$
	strategy execution coefficient	$l = 10$
HFMOEA [67]	crossover rate	$CR = 0.9$
	mutation rate	$MR = 1/D$
	filter rate	$Fr = 50\%$
	filter initialized individual proportion	$Finit = 0.1$
MOPSO-ASFS [48]	inertia weight	$\omega = 0.7298$
	learning rate	$c1 = c2 = 1.46$
	maximum velocity	$vmax = 4$
	minimum velocity	$vmin = -4$
CMODE [59]	crossover factor	$Cr = 0.5$
	mutation factor	$F = 0.5$
	size of leader set	$\alpha = 5$
MOGS-MLPSAE	crossover rate of learning high-level pool	$CR_1 = 0.9$
	mutation factor	$MR = 1/D$
	decay factor	$\alpha = 0.15$
	dimensional limitation	dim = 200
Shared Parameters	maximum number of iterations	maxIN = 100
	population size	$N = 100$
	number of experimental independent runs	$T = 30$
	number of nearest neighbors	$k = 3$
	number of filter features	$rf = 1000$
	number of nearest neighbors of reliefF	$Rk = 5$

5) **Population Update**: Store parent individual (PI) and offspring individuals (OIL) (line 12).
6) **Population Expansion**: Repeat Steps 4–5 until the population size reaches $2N$, where N represents the number of individuals in the initialized population.
7) **Iteration**: Repeat Steps 3–6 until the maximum number of iterations is reached.
8) **Optimal Subset Selection**: Select the optimal gene subset from population P based on the classification performance of the subset.

The complexity of MOGS-MLPSAE primarily resides in the wrapper phase, which corresponds to steps 2–7 of Algorithm 4. During the algorithm's search for the optimal solution set, the population is iterated maxIN times. The complexity of the algorithm at each iteration is influenced by several factors: population initialization $O(N)$ in step 2, the non-dominated sorting $O(N^2)$ and PRPD strategy $O(2N)$ in step 3, as well as the process of cyclic iteration utilizing the PBE strategy to generate the offspring individuals in steps 4, 5, and 6. Specifically, generating these $2N$ individuals involves ensuring that each individual adheres to different rules (Eqs.5–9). Each individual produces a random number of offspring (1−5), averaging 3. Consequently, the number of iterations is constant and independent of N, leading to a per-iteration complexity of $O(N)$. Given that the number of iterations in the entire evolutionary process is maxIN, the overall computational complexity of MOGS-MLPSAE is $O(\max IN \times N^2)$.

Algorithm 3. Population Initialization

Input: $D_{s\times t}$(dataset with s samples and t features); dim(dimensional limit of the space for initializing the population); N(the number of population individuals);
Output: P (population);
initialize $P[p_i] = 1 \times t$ matrix of zeros for $i = 1$ to N;
$FList = \{1, 2, 3, 4, \ldots, t\}$;
for each individual p_i in P **do**
  randomly select dim values from $FList$ to generate a list Rf;
  generate a random number r between 1 and dim;
  randomly select r values from Rf to generate a list R;
  for $j = 1$ **to** $|R|$ **do**
    p_{i,R_j} =1; // p_{i,R_j} represents the R_jth feature of the ith individual of the population P
  end
end
Return: P;

Algorithm 4. The framework of MOGS-MLPSAE

Input: $D_{s\times t}$ (dataset with s samples and t features); $maxIN$ (the maximum iteration number); dim (dimensional limit of the space for initializing the population); N (the number of population individuals);
Output: Ogs (the optimal gene subsets);
generate a reduced dataset $D_{s\times t'}$ by selecting the top 5% of features with ReliefF;
initialize the population P in the dim-dimensional space of $D_{s\times t'}$ based on **Algorithm 3**;
$nIN = 0$;
while $nIN < maxIN$ **do**
  non-dominated sorting for P based on $D_{s\times t'}$ and Eq.2;
  get the divided rank pool RP based on **Algorithm 1**;
  $TP = \{\}$; // Initialize an empty population
  $TPcount = 0$;
  while $TPcount < 2N$ **do**
    select a parent individual PI based on Pareto rank and unit pool rank order;
    parent individual PI produces m offspring individuals OIL based on the traditional mutation operator and **Algorithm 2**;
    save PI and OIL individuals to TP;
    $TPcount = TPcount + 1 + m$;
  end
  $P = TP$;
end
select the optimal gene subset Ogs from population P;
Return: Ogs;

In the initialization phase of the MOGS-MLPSAE, the PRPD strategy constructs the algorithm based on a ranking unit pool. This facilitates individual guidance and enhances algorithmic efficiency. The PRPD strategy assigns each individual both a Pareto front rank and a unit pool rank. Individuals in high-level pools exhibit higher accuracy and can guide individuals in low-ranked pools, thereby aiding them in improving their accuracy. This approach ensures accurate identification of the role of each individual and enables more effective guidance for enhancing algorithm performance.

In the evolutionary phase of the MOGS-MLPSAE, the PBE strategy utilizes five new rules to guide the unit pools, precisely directing

Table 4
The average and standard deviation of the classification accuracies (%) from 8 algorithms across 14 datasets over 30 independent runs.

Dataset	NSGA-II	NSGA-III	HFMOEA	MOBGA-AOS	GR-MOEA	MOPSO-ASFS	CMODE	MOGS-MLPSAE
TOX_171	83.99 ± 0.66	85.63 ± 0.59	85.33 ± 1.02	83.17 ± 0.64	87.67 ± 1.53	85.06 ± 0.38	85.45 ± 1.18	**89.23 ± 1.93**
Leukemia	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Arcene	93.86 ± 1.19	95.03 ± 0.97	96.03 ± 1.34	95.60 ± 1.28	94.22 ± 0.89	94.72 ± 0.92	95.97 ± 0.00	**97.38 ± 0.49**
Brain_Tumor	88.77 ± 1.46	90.83 ± 1.06	90.49 ± 1.13	93.84 ± 1.37	91.60 ± 1.21	89.65 ± 1.07	92.77 ± 0.37	**94.40 ± 1.50**
CLLSUB	84.78 ± 1.63	86.09 ± 0.94	86.63 ± 0.98	86.26 ± 1.05	87.44 ± 1.41	85.62 ± 0.41	85.02 ± 0.13	**92.82 ± 1.84**
11_Tumor	93.92 ± 1.35	94.22 ± 0.76	94.54 ± 0.95	**95.27 ± 0.36**	94.03 ± 0.53	94.85 ± 0.56	95.13 ± 0.25	94.87 ± 1.33
LungCancer	98.87 ± 0.54	99.17 ± 0.96	99.35 ± 0.15	99.42 ± 0.15	**100.00 ± 0.00**	99.67 ± 0.27	**100.00 ± 0.00**	**100.00 ± 0.00**
Prostate	95.74 ± 1.02	97.23 ± 0.20	96.90 ± 0.56	97.57 ± 0.32	97.79 ± 0.25	96.54 ± 0.51	98.00 ± 0.00	**98.90 ± 0.70**
Ovarian	97.86 ± 0.42	98.76 ± 0.86	98.45 ± 0.12	98.34 ± 0.09	98.33 ± 0.12	98.45 ± 0.13	98.87 ± 0.01	**100.00 ± 0.00**
SMKCAN187	83.86 ± 0.82	85.93 ± 0.59	84.76 ± 1.24	83.55 ± 0.95	88.50 ± 1.34	83.86 ± 0.67	84.89 ± 0.01	**90.45 ± 1.18**
BreastCancer2	98.04 ± 1.19	99.18 ± 1.24	98.48 ± 1.05	98.20 ± 1.32	98.95 ± 0.77	98.04 ± 0.83	99.10 ± 1.11	**100.00 ± 0.00**
CrohnDisease	95.77 ± 1.06	97.26 ± 1.26	96.32 ± 0.12	96.39 ± 0.56	96.41 ± 0.38	95.77 ± 0.29	97.18 ± 0.12	**98.51 ± 0.74**
GLI	96.91 ± 0.93	98.02 ± 0.63	97.81 ± 0.75	98.03 ± 0.81	98.20 ± 0.33	96.91 ± 0.27	98.75 ± 1.12	**100.00 ± 0.00**
BreastCancer1	92.97 ± 0.75	94.20 ± 0.74	93.21 ± 0.67	93.48 ± 0.75	94.06 ± 0.48	92.97 ± 0.64	92.78 ± 0.03	**96.21 ± 0.64**
+ / − / =	13/0/1	11/0/3	11/0/3	11/1/2	10/0/4	11/0/3	9/1/4	
Ranking	6.93	3.86	4.64	4.43	3.57	5.57	3.21	**1.14**

Table 5
The number of features selected by eight algorithms across 14 datasets.

Dataset	NSGA-II	NSGA-III	HFMOEA	MOBGA-AOS	GR-MOEA	MOPSO-ASFS	CMODE	MOGS-MLPSAE
TOX_171	129.57	81.41	99.23	57.65	92.37	63.13	47.90	**27.80**
Leukemia	94.99	47.42	56.59	19.96	78.08	25.40	14.06	**3.70**
Arcene	173.58	74.54	105.32	63.27	88.82	74.33	60.00	**25.60**
Brain_Tumor	106.44	58.03	60.92	66.02	73.75	32.00	20.96	**8.40**
CLLSUB	174.52	57.64	116.85	75.33	89.74	72.92	54.63	**14.80**
11_Tumor	115.88	48.86	104.15	146.20	136.24	117.42	114.23	**34.10**
LungCancer	144.20	74.41	97.33	63.01	89.74	62.90	48.21	**3.70**
Prostate	101.13	48.08	87.12	96.15	107.68	75.23	66.81	**6.30**
Ovarian	109.39	73.62	117.25	100.87	77.68	73.54	55.28	**2.80**
SMKCAN187	199.07	116.85	148.82	155.58	123.45	114.05	96.06	**20.30**
BreastCancer2	76.23	59.87	63.14	69.56	42.31	55.92	50.17	**5.30**
CrohnDisease	107.62	57.64	97.73	137.81	65.11	93.91	86.85	**24.90**
GLI	142.54	77.55	101.26	152.22	84.63	117.02	114.63	**7.30**
BreastCancer1	157.41	76.37	130.08	120.26	114.23	142.66	137.68	**23.80**

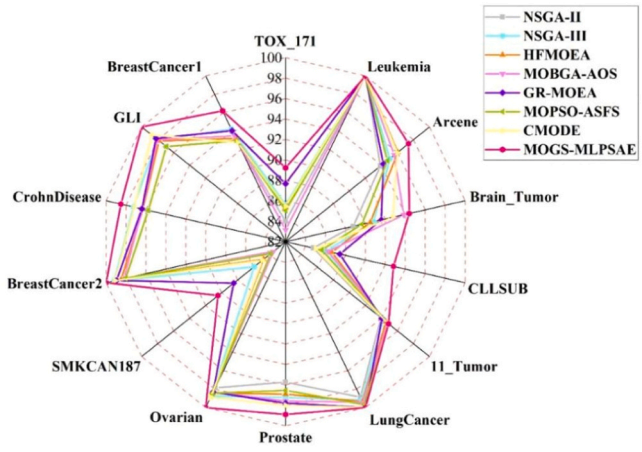


Fig. 8. Comparison of *Avg_Acc* on 14 datasets.

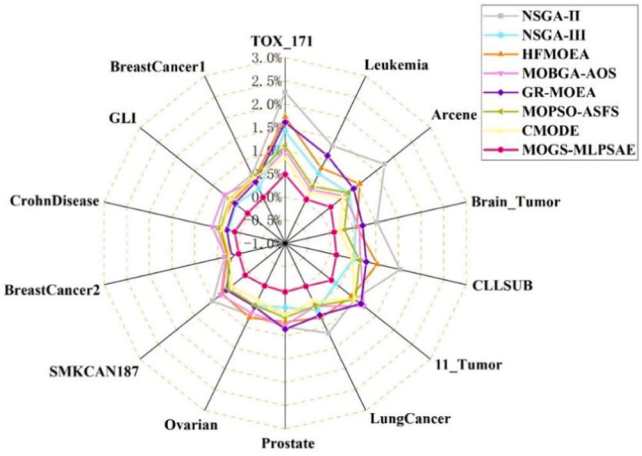


Fig. 9. Comparison of gene selection rate on 14 datasets.

individuals of different ranks for selective cross-rank learning. Serving as the core search mechanism of the MOGS-MLPSAE algorithm, this strategy enhances exploration and exploitation, thereby achieving high classification accuracy while maintaining population diversity. By prioritizing high accuracy as the primary objective of the algorithm, the PBE strategy enables more effective discovery of a set of individuals with high accuracy. This effectively addresses the challenge in traditional multi-objective optimization algorithms of balancing two objectives without clear prioritization, making a significant contribution to the improved application of multi-objective optimization algorithms in feature selection.

Additionally, the PAE mechanism is proposed based on the PRPD and PBE strategies. As a new population evolution mechanism, it adaptively determines the number of parent individuals selected according to the current population size. This ensures the continuous advancement of the population. The adaptive evolution mechanism enhances both the robustness and generalization capability of the MOGS-MLPSAE algorithm.

4. Experimental results

In this section, the performance of the MOGS-MLPSAE algorithm and seven comparative algorithms is evaluated using 14 public benchmark microarray datasets. These datasets were exclusively obtained from publicly accessible databases, specifically the Genomic Data Commons Data Portal (https://portal.gdc.cancer.gov/) and the Xena Browser (https://xenabrowser.net/datapages/). The experimental settings are described in Section 4.1. Subsequently, the results of the comparison experiments are analyzed in Section 4.2. Furthermore, the effectiveness of the PBE mechanisms is assessed through an ablation experiment in Section 4.3. Finally, the sensitivity analysis of parameters is presented in Section 4.4. All experiments were carried out on an Intel Core-i5-6500 CPU 3.20 GHZ with 8 GB RAM, using Python implementation.

4.1. Experimental setting

4.1.1. Datasets

Fourteen public benchmark microarray datasets were selected in our experiment, as presented in Table 2 along with their basic information. All datasets exhibit high dimensionality and low sample size, with the number of features (genes) ranging from 5748 to 24481 and the number of samples ranging from 50 to 252. Additionally, these datasets comprise nine binary classification datasets and five multi-class datasets.

4.1.2. Comparison algorithms and parameter settings

In our experiment, seven competitive multi-objective FS algorithms were selected to compare their performance with the proposed MOGS-MLPSAE. These algorithms include NSGA-II [48], NSGA-III[65], MOBGA-AOS [57], GR-MOEA [66], HFMOEA [67], MOPSO-ASFS [59] and CMODE [68]. All seven compared algorithms have demonstrated effectiveness in solving a variety of MOOPs. To ensure the validity of the comparative studies, the parameters of these algorithms were fine-tuned, aligning them with the recommendations provided by their respective authors. Table 3 presents a comprehensive overview of the parameter settings of all comparison algorithms.

NSGA-II, NSGA-III, MOBGA-AOS, GR-MOEA, and HFMOEA all utilize classical crossover and mutation operators, maintaining consistent crossover rates (*CR*) and mutation rates (*MR*). The proposed MOGS-MLPSAE algorithm retains the traditional mutation operator and introduces a self-attenuating crossover operator with an initial crossover

Table 6
Comparison of Macro F1-Score across 14 datasets.

Dataset	NSGA-II	NSGA-III	HFMOEA	MOBGA-AOS	GR-MOEA	MOPSO-ASFS	CMODE	MOGS-MLPSAE
TOX_171	0.6999	0.7282	0.7194	0.6864	0.7552	0.7203	0.7201	**0.7789**
Leukemia	**1.0000**	**1.0000**	**100.00**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Arcene	0.8698	0.8939	0.9093	0.9017	0.8793	0.8884	0.9210	**0.9435**
Brain_Tumor	0.7780	0.8153	0.8086	**0.8769**	0.8279	0.7941	0.8571	0.8677
CLLSUB	0.7049	0.7330	0.7419	0.7350	0.7522	0.7295	0.7217	**0.8445**
11_Tumor	0.8694	0.8805	0.8847	**0.9042**	0.8791	0.8943	0.9025	0.8874
LungCancer	0.9721	0.9739	0.9855	0.9869	**1.0000**	0.9907	**1.0000**	**1.0000**
Prostate	0.9068	0.8850	0.9335	0.9488	0.9538	0.9270	**0.9712**	0.9604
Ovarian	0.9535	0.9669	0.9680	0.9661	0.9556	0.9679	0.9774	**1.0000**
SMKCAN187	0.6963	0.7333	0.7079	0.6901	0.7713	0.6976	0.7205	**0.8074**
BreastCancer2	0.9495	0.9713	0.9594	0.9513	0.9714	0.9530	0.9710	**1.0000**
CrohnDisease	0.9070	0.9336	0.9265	0.9237	0.9258	0.9144	0.9432	**0.9631**
GLI	0.9301	0.9546	0.9493	0.9530	0.9610	0.9365	0.9641	**1.0000**
BreastCancer1	0.8573	0.8804	0.8625	0.8669	0.8802	0.8584	0.8605	**0.9194**

rate (CR_1) of 0.9 and a decay factor (α) of 0.15. During the population initialization of MOGS-MLPSAE, the number of features selected by each individual is restricted to no more than 200. NSGA-III improves the selection mechanism by employing a reference point method with a reference point number (*Pnum*) of 4. MOBGA-AOS integrates five operators with different search capabilities, each of which is used alternately and repeated five times ($LP = 5$). GR-MOEA is an interactive filter- wrapper multi-objective evolutionary algorithm that constructs both filter and wrapper populations to co-evolve, interacting with each other every 10 generations (l). The HFMOEA algorithm includes an independent filter with a filter rate (*Fr*) of 0.9 and a filtering proportion (*Finit*) of 0.1 for the initialized population individuals. MOPSO-ASFS is an improved particle swarm optimization algorithm (PSO) with parameters such as inertia weight (ω), learning rate ($c1$, $c2$), maximum velocity (*vmax*), and minimum velocity (*vmin*). CMODE is an efficient multi-objective differential evolution algorithm designed with a new shifted density estimation mutation strategy to balance population diversity and convergence during the evolutionary process. The leader set (α) selects superior individuals to guide population evolution, with F and Cr representing the mutation rate and crossover rate of the differential evolution algorithm.

In addition, all algorithms were independently executed 30 times on each dataset to mitigate the impact of random variations. The KNN [69] was used to evaluate feature subsets (with $k = 3$), and five-fold cross--validation was employed to assess the classification accuracy. Moreover, all algorithms utilized the ReliefF filtering process to ensure fairness, with the default value of 5 being set as the number of nearest neighbors for a sample selected by ReliefF.

4.2. Comparison of experiment results and analysis

A series of experiments were conducted to illustrate the effectiveness and efficiency of the proposed approach. Wilcoxon rank-sum test, with a significance level of 0.05, was adopted to evaluate the performance of the proposed feature selection method with 30 independent runs. In the tables, the symbol " + ", " − " and "≈" indicate that MOGS-MLPSAE is significantly better than, worse than, and of no significant difference to the other algorithms, respectively. In addition, the Pareto front of each algorithm is composed of all non-dominated solutions in 30 independent runs.

4.2.1. Analysis of the classification performance

The effectiveness of feature selection (FS) methods is commonly evaluated based on two crucial metrics: classification accuracy and the number of selected features. In this experiment, an algorithm that demonstrates superiority in both metrics effectively addresses gene selection problems, thereby illustrating its efficacy. Furthermore, classification accuracy remains the primary metric for evaluating the algorithm's overall classification performance.

Table 4 shows the average (*Avg_Acc*) and standard deviation (*Std*) of the classification accuracies of MOGS-MLPSAE and the other seven compared algorithms, based on the 30 independent runs conducted on the 14 microarray datasets listed in Table 2. The bold values indicate that the corresponding method obtains the best result. According to the results of the experiments, MOGS-MLPSAE achieves excellent classification accuracy compared to the other methods on 13 out of the 14 datasets. Although MOGS-MLPSAE does not achieve the highest classification accuracy on the 11_Tumor dataset, it still ranks third and exhibits a leading advantage in another indicator (as shown in Table 5). The MOGS-MLPSAE algorithm achieves a remarkable classification accuracy of 96.21 % for BreastCancer1, surpassing the other seven compared algorithms by at least 2.01 %. The comparative algorithms failed to achieve a classification accuracy breakthrough of 90 % for the TOX_171, CLLSUB, and SMKCAN187 datasets, indicating their relative complexity and challenge. However, MOGS-MLPSAE outperformed all other algorithms with a lead ranging from 1.56–8.04 % in classification accuracy. Moreover, while all algorithms achieved similar accuracy for the Leukemia and LungCancer datasets, MOGS-MLPSAE demonstrated the capacity to select a smaller average number of genes.

The statistical results of the Wilcoxon rank sum test, summarized in the penultimate row of Table 4, indicate that MOGS-MLPSAE outperforms other algorithms on most datasets. The last row gives the algorithm rankings based on the Friedman test [70]. Notably, MOGS-MLPSAE ranked first in the comprehensive ranking of average classification accuracy across all datasets, demonstrating its significantly superior performance compared to other comparative algorithms. To provide a more visual representation of the classification performance of the comparison algorithms, Fig. 8 shows a radar chart depicting the *Avg_Acc* indicator for each algorithm across all datasets.

The number of selected features based on the 30 independent runs on the 14 microarray datasets is compared between MOGS-MLPSAE and the other seven algorithms, as presented in Table 5. From the results of the experiments, the MOGS-MLPSAE algorithm selected fewer genes than other comparison algorithms across all datasets. Although MOGS-MLPSAE prioritizes high classification accuracy as its main goal, it still achieves advantages in the number of gene selections. This implies that the gene subset with high classification accuracy can contain fewer genes. Consequently, MOGS-MLPSAE effectively reduces the number of genes while pursuing high classification accuracy, thereby substantiating the efficacy of the population-biased evolution (PBE) strategy. To intuitively demonstrate the performance of the algorithm in terms of the number of selected features, Fig. 9 presents a radar chart illustrating the FS rate for each algorithm across all datasets. The radar chart indicates that MOGS-MLPSAE exhibits the lowest gene selection rate on all datasets, with an average rate of only 1 % and a minimum of merely 0.01 %. This highlights MOGS-MLPSAE's significant advantage in gene selection, effectively preserving valuable information from the original data while reducing its dimensions.

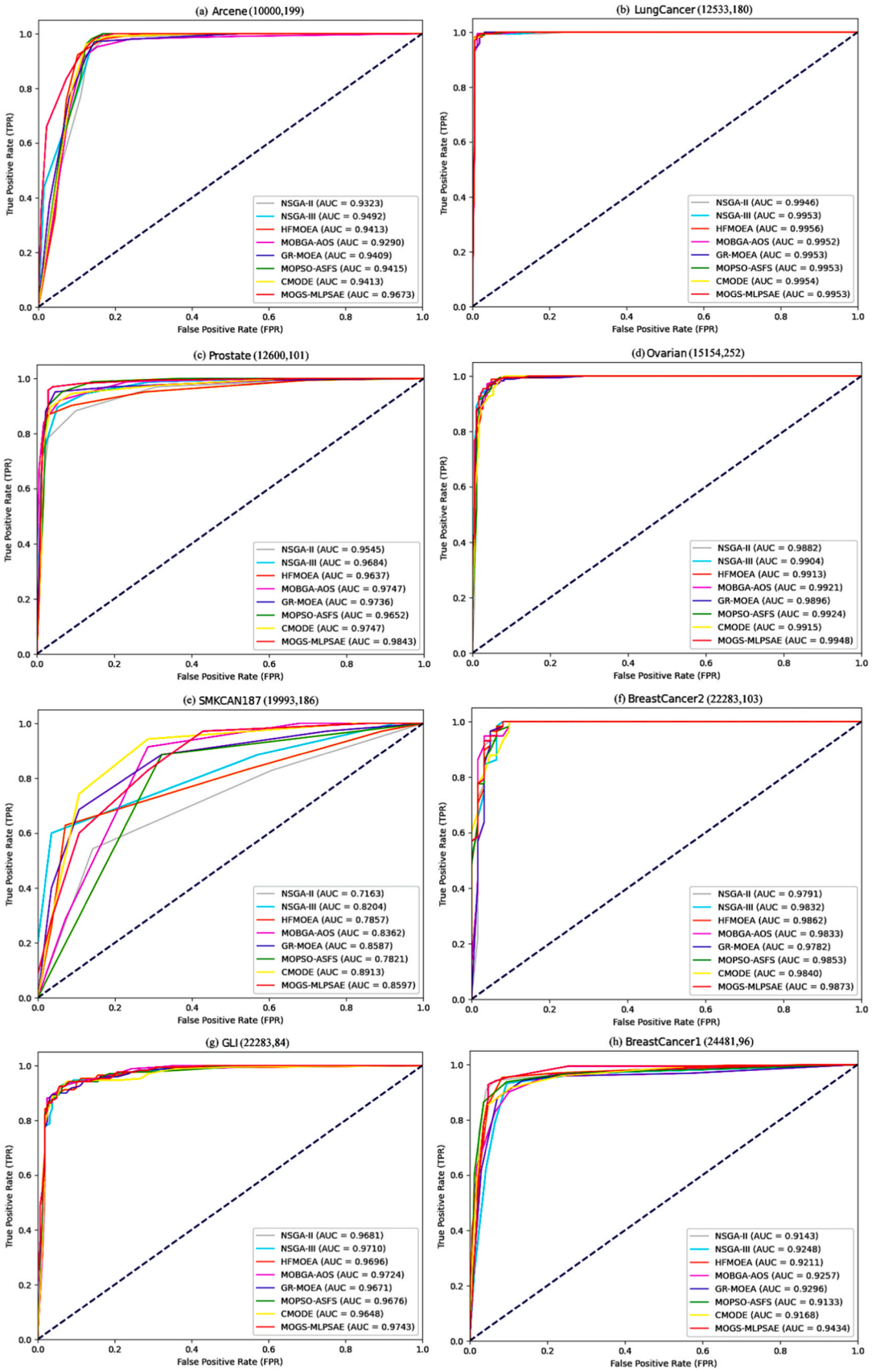


Fig. 10. Comparison of ROC curves.

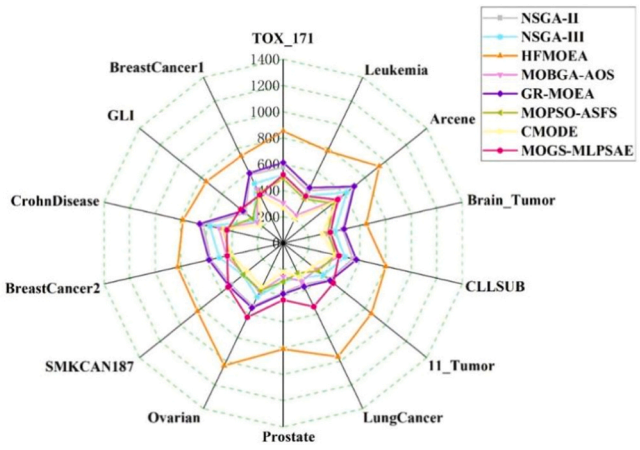


Fig. 11. Comparison of running time across 14 datasets.

To provide a more comprehensive evaluation of the classification performance of the proposed MOGS-MLPSAE algorithm, we have incorporated the Macro F1-score as an additional assessment metric (Table 6) and included ROC curve analysis (Fig. 10). The F1-score, defined as the harmonic mean of precision (P) and recall (R), serves as a balanced measure of the classification model for a class. A higher F1-score indicates a superior balance between precision and recall in a classification model. The Macro F1-score is calculated by taking the arithmetic mean of the F1-scores of each class. As shown in Table 6, MOGS-MLPSAE achieved the highest Macro F1-score in 11 of the 14 datasets, underscoring its substantial advantages. Furthermore, Fig. 10 presents the ROC curve comparison among eight algorithms across eight binary classification datasets, which also underscores the comparative superiority of the MOGS-MLPSAE algorithm. In summary, the proposed MOGS-MLPSAE demonstrates superior overall performance relative to other comparison algorithms.

4.2.2. Analysis of the time costs

Fig. 11 illustrates a radar chart comparing the runtime of eight algorithms across 14 datasets. The chart shows that MOGS-MLPSAE exhibits consistently stable performance in terms of runtime, with no significant fluctuations. This stability suggests that MOGS-MLPSAE possesses robust adaptability to diverse datasets. Although MOGS-MLPSAE does not have the shortest runtime, its maximum runtime remains within 30 min. Considering that gene selection for microarray data is a preprocessing step in constructing classification models, the computational time is reasonable and justifiable.

4.2.3. Analysis of the Pareto Front

The MOGS-MLPSAE framework is an enhanced Pareto-based multi-objective search approach designed to address the challenge faced by traditional MOOAs in improving the search for high classification accuracy objectives in gene selection. To validate the effectiveness of the proposed algorithm, Fig. 12 illustrates the Pareto front of MOGS-MLPSAE and seven comparison algorithms across eight datasets. Each subgraph corresponds to a dataset, with the vertical axis showing classification error rate and the horizontal axis showing the number of selected genes. It should be noted that a scatter distribution closer to the lower-left corner indicates superior performance of the algorithm in searching for gene subsets. The number of genes and samples in the original dataset is also labeled at the top of each subgraph for reference.

As depicted in Fig. 12, it is evident that the MOGS-MLPSAE algorithm exhibits significant advantages on seven out of the eight datasets. Although it does not achieve optimal performance in terms of classification error rate for the 11_Tumor dataset, MOGS-MLPSAE still performs competitively. More importantly, it also demonstrates a leading advantage in terms of the size of the selected genes. From the perspective of the algorithm's scatter distribution, the scatter of the MOGS-MLPSAE algorithm is closer to the vertical axis. In other words, the population individuals of MOGS-MLPSAE algorithm are closer to the smaller number of genes. This indicates that the MOGS-MLPSAE algorithm can effectively reduce the number of selected genes while maintain high classification accuracy as the primary guiding goal.

4.3. Analysis of PBE strategy

The PBE strategy incorporates five new rules to constrain the evolutionary behavior of individuals in the unit pool, guiding the population's biased evolution, which serves as the core search mechanism of MOGS-MLPSAE. To optimize the effectiveness of this strategy, we devised five variants of MOGS-MLPSAE (MOGS-MLPSAE-I, MOGS-MLPSAE-II, MOGS-MLPSAE-III, MOGS-MLPSAE-IV, and MOGS-MLPSAE-V) to assess the impact of adopting different rules within the PBE strategy. Each variant of the PBE strategy omits one of these five rules. For instance, rule 1 (Eq.5) is absent in MOGS-MLPSAE-I; rule 2 (Eq.6) is omitted in MOGS-MLPSAE-II; rule 3 (Eq.7) is not included in MOGS-MLPSAE-III; rule 4 (Eq.8) is excluded from MOGS-MLPSAE-IV, and finally, rule 5 (Eq.9) is disregarded in case of MOGS-MLPSAE-V. The average classification accuracy (*Avg_Acc*) and the number of selected genes (*#Genes*) for MOGS-MLPSAE and its five variants across eight datasets are presented in Table 7.

As shown in Table 7, MOGS-MLPSAE-II exhibits the poorest results across the eight datasets. This indicates the paramount importance of rule 2 in enhancing the algorithm's performance; in other words, it plays a pivotal role in augmenting the effectiveness of the PBE strategy. The contribution of rule 2 can be attributed to its capacity to optimize the learning of the unit pool across levels. Additionally, the classification performance of MOGS-MLPSAE-V is inferior to all other algorithms except MOGS-MLPSAE-II. This finding highlights the substantial contribution of incorporating individuals with high classification accuracy guided by rule 5 in enhancing the algorithm's classification performance. Furthermore, MOGS-MLPSAE outperforms all other variants across the eight datasets. This demonstrates that each rule is indispensable and contributes to the PBE strategy to improve algorithm performance.

4.4. Sensitivity analysis of parameters

The parameter sensitivity analyses of the crossover rates CR_1, CR_2, and CR_3 for learning high-level, mid-level, and low-level unit pools, respectively, as well as the decay factor α and the dimensional limit dim of the space for initializing the population, are conducted on three representative datasets: BreastCancer1, Brain_Tumor, and CLLSUB. These datasets were chosen because they contain a larger number of genes and categories but fewer samples, making them more challenging for classification tasks.

Leveraging Taguchi's experimental design [71] with its orthogonal array for efficient analysis, we identified a suitable combination of three parameters. This method minimized the number of experiments required by analyzing various parameter combinations simultaneously. The levels for three parameters are set as follows: five levels for $CR_1 \in \{0.30, 0.45, 0.60, 0.75, 0.90\}$, five levels for $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$, and five levels for dim $\in \{50, 100, 200, 300, 400\}$. Therefore, an orthogonal array $L_{25}(5^3)$ was used, which comprises 25 experiments as enumerated in Table 8. The *Acc_Mean* and *#Genes* were used to evaluate the MOGS-MLPSAE with 25 parameter combinations.

The results of 25 parameter combinations on three datasets are ranked according to *Acc_Mean* metric, as the primary objective of FS is to improve classification performance. Table 9 presents the top 10 parameter combinations, with the best values highlighted in bold. From Tables 8 and 9, the parameter combination {0.90, 0.15, 200} performs best in terms of *Acc_Mean* metrics across the three datasets. Although it

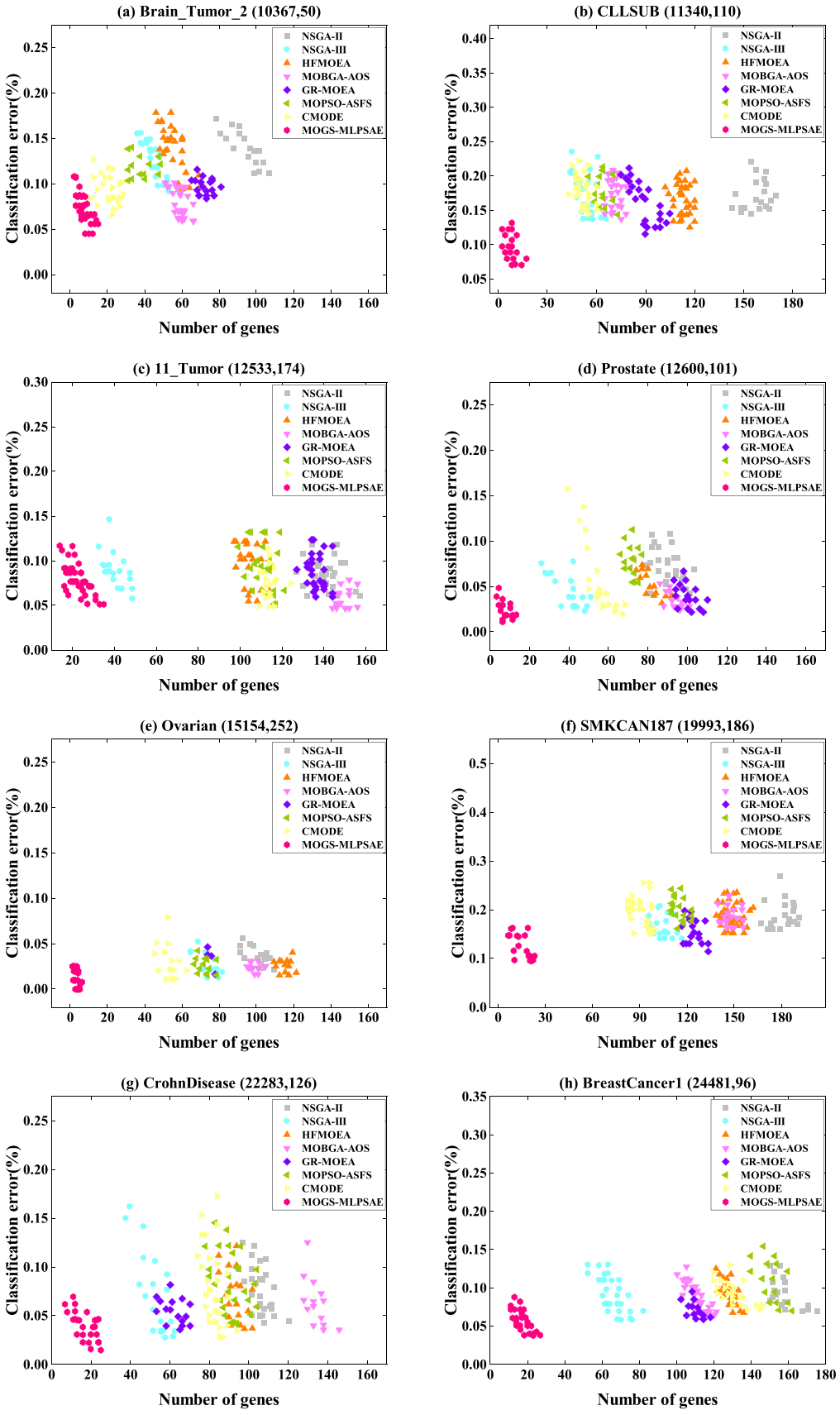


Fig. 12. Comparison of Pareto fronts.

Table 7
Avg_Acc and *#Genes* of MOGS-MLPSAE and its five variants on 8 datasets.

Dataset	Method	*Acc_Mean*	*# Genes*
Brain_Tumor	MOGS-MLPSAE-I	93.19	25.40
	MOGS-MLPSAE-II	92.06	36.20
	MOGS-MLPSAE-III	93.81	10.30
	MOGS-MLPSAE-IV	94.12	19.60
	MOGS-MLPSAE-V	92.77	34.60
	MOGS-MLPSAE	**94.40**	**8.40**
CLLSUB	MOGS-MLPSAE-I	91.87	35.20
	MOGS-MLPSAE-II	90.39	47.70
	MOGS-MLPSAE-III	92.27	16.50
	MOGS-MLPSAE-IV	92.55	24.70
	MOGS-MLPSAE-V	91.23	44.80
	MOGS-MLPSAE	**92.82**	**14.80**
11_Tumor	MOGS-MLPSAE-I	93.27	47.20
	MOGS-MLPSAE-II	92.32	59.30
	MOGS-MLPSAE-III	94.18	34.90
	MOGS-MLPSAE-IV	94.59	42.30
	MOGS-MLPSAE-V	92.63	56.50
	MOGS-MLPSAE	**94.87**	**34.10**
Prostate	MOGS-MLPSAE-I	97.59	24.70
	MOGS-MLPSAE-II	96.87	33.50
	MOGS-MLPSAE-III	98.24	8.50
	MOGS-MLPSAE-IV	98.56	15.40
	MOGS-MLPSAE-V	97.23	29.20
	MOGS-MLPSAE	**98.90**	**6.30**
Ovarian	MOGS-MLPSAE-I	99.15	12.50
	MOGS-MLPSAE-II	98.24	17.80
	MOGS-MLPSAE-III	99.43	5.60
	MOGS-MLPSAE-IV	99.65	8.90
	MOGS-MLPSAE-V	98.77	13.20
	MOGS-MLPSAE	**100.00**	**2.80**
SMKCAN187	MOGS-MLPSAE-I	89.26	35.20
	MOGS-MLPSAE-II	88.05	46.80
	MOGS-MLPSAE-III	89.57	22.10
	MOGS-MLPSAE-IV	89.96	28.50
	MOGS-MLPSAE-V	88.43	41.30
	MOGS-MLPSAE	**90.45**	**20.30**
CrohnDisease	MOGS-MLPSAE-I	97.52	39.70
	MOGS-MLPSAE-II	96.74	57.60
	MOGS-MLPSAE-III	97.89	27.30
	MOGS-MLPSAE-IV	98.27	32.60
	MOGS-MLPSAE-V	97.18	51.50
	MOGS-MLPSAE	**98.51**	**24.90**
BreastCancer1	MOGS-MLPSAE-I	95.23	41.50
	MOGS-MLPSAE-II	94.47	53.60
	MOGS-MLPSAE-III	95.56	26.60
	MOGS-MLPSAE-IV	95.89	29.40
	MOGS-MLPSAE-V	94.87	48.20
	MOGS-MLPSAE	**96.21**	**23.80**

Table 8
Parameter combinations of experiments.

No.	CR_1	α	dim	No.	CR_1	α	dim
1	0.90	0.05	50	14	0.60	0.20	50
2	0.90	0.10	100	15	0.60	0.30	100
3	0.90	0.15	200	16	0.45	0.05	300
4	0.90	0.20	300	17	0.45	0.10	400
5	0.90	0.30	400	18	0.45	0.15	50
6	0.75	0.05	100	19	0.45	0.20	100
7	0.75	0.10	200	20	0.45	0.30	200
8	0.75	0.15	300	21	0.35	0.05	400
9	0.75	0.20	400	22	0.35	0.10	50
10	0.75	0.30	50	23	0.35	0.15	100
11	0.60	0.05	200	24	0.35	0.20	200
12	0.60	0.10	300	25	0.35	0.30	300
13	0.60	0.15	400				

may not achieve optimal results in terms of *#Genes* metrics, its performance remains comparable. Consequently, we set CR_1 to 0.90, α to 0.15, and dimension dim to 200 as these values represent an ideal combination.

5. Conclusion and future work

In this study, we propose a novel multi-objective gene selection method, MOGS-MLPSAE, based on a multi-level pool and self-adaptive evolution. This method enhances the guiding role of individuals with high classification accuracy through several key contributions: (1) a Pareto-based ranking pool division strategy that facilitates cross-level learning and improves evolutionary guidance; (2) a population-biased evolutionary mechanism with five rules to ensure the population evolves toward higher classification accuracy; and (3) a self-adaptive evolution mechanism to balance the number of parent and offspring individuals, addressing the challenge of unspecified population sizes. Comparative evaluations of MOGS-MLPSAE against seven competitive multi-objective algorithms across 14 microarray datasets demonstrate its superior classification accuracy. For instance, on the BreastCancer1 dataset, MOGS-MLPSAE outperformed other algorithms by at least 2.01 %, achieving an accuracy of 96.21 %. Furthermore, across all 14 datasets, MOGS-MLPSAE consistently selected a smaller number of genes compared to the competing algorithms, with an average gene selection rate of only 1 % and a minimum of 0.01 %.

While MOGS-MLPSAE shows promising performance, it is important

Table 9
Result of the top 10 parameter combinations on 3 datasets.

Rank	BreastCancer1			Brain_Tumor			CLLSUB		
	No.	*Acc_Mean*	# Genes	No.	*Acc_Mean*	# Genes	No.	*Acc_Mean*	# Genes
1th	3	**96.21**	23.8	3	**94.4**	8.4	3	**92.82**	14.8
2th	2	96.1	21.8	4	94.38	7.4	6	92.73	15.8
3th	4	95.96	**19.8**	7	94.21	9.4	17	92.59	**3.8**
4th	7	95.91	31.8	2	94.17	**2.4**	4	92.55	16.8
5th	11	95.56	29.8	5	94.02	21.4	7	92.13	26.8
6th	6	95.21	25.8	6	93.46	31.4	11	91.48	13.8
7th	5	94.64	39.8	11	92.71	5.4	5	91.07	5.8
8th	8	94.29	45.8	20	92.08	35.4	4	90.55	18.8
9th	16	93.31	22.8	24	91.23	6.4	3	90.33	30.8
10th	20	93.01	50.8	2	90.58	40.4	13	89.9	43.8

to address its limitations and outline future improvements. A key limitation is the need to further explore and optimize its self-adaptive evolution mechanism for balancing parent and offspring individuals. Future improvements could focus on refining the ranking pool division strategy to better adapt to various dataset characteristics and structures. Continuous refinement of the rules governing individual evolutionary behavior is essential to fine-tune the algorithm's performance. Moreover, exploring methods to enhance the scalability of the algorithm to handle larger datasets and improve computational efficiency would significantly contribute to its broader practical applicability.

CRediT authorship contribution statement

Xiang Yu: Formal analysis, Conceptualization. **Shaobo Deng:** Formal analysis, Conceptualization. **Junke Wang:** Data curation, Conceptualization. **Chen Jin:** Investigation, Data curation. **Rutun Cao:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Min Li:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Research on this work is partially supported by grants from the National Natural Science Foundation of China (No.62166028).

Data availability

Data will be made available on request.

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